9:27 79%

# AI Forecast

● LSTM model · 60 simulated readings

24 Hours | 7 Days

Demand forecast (Wh)

LSTM active

Mon Thu Sun Wed

Predicted Confidence band

| 28.3 W | 27.0 W | 170 Wh |
|---|---|---|
| Peak predicted | Avg generation | Est. daily |

**AI Recommendation**

High generation avg 27.0W. Peak window 10AM-2PM. Schedule relay load during peak. Battery estimated 60% – excellent for charging.

Hourly prediction · peak 28.3W

| | |
|---|---|
| 6 AM | 6.2 Wh |
| 9 AM | 18.4 Wh |
| 12 PM | 27.7 |

Dashboard | Forecast | Load | Battery | Hardware

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Hardware & Data
ESP32 · IoT prototype · Firebase RTDB
Records collected
60 readings
Wi-Fi RSSI
-73 dBm
ESP32 uptime
106s
Update interval
Every 10 seconds
Firebase project
solargrid-ai-1c048
App writes to
relay_command/state only
Data source
Mock simulation
Wiring reference
GPIO2
Relay IN
GPIO21
INA219 SDA (I²C)
GPIO22
INA219 SCL (I²C)
GPIO34
Voltage sensor S
GPIO35
ACS712 OUT (÷2)
VIN
LM2596 OUT+ (5V)
3.3V
INA219 VCC
GND
Common ground
Dashboard
Forecast
Load
Battery
Hardware

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# Hardware & Data

ESP32 · IoT prototype · Firebase RTDB

Demo mode · 60 simulated readings · ESP32 offline

## Sensors · live values

### INA219 Current/Voltage

I²C · GPIO21 (SDA) · GPIO22 (SCL) · 0x40 — OK

12.88V · 2000mA · 27.8W

### ACS712 Current sensor

GPIO35 · 10kΩ+10kΩ voltage divider · 20A module — OK

1935 mA

### Voltage sensor (12V)

GPIO34 · analog ADC · ÷5 ratio · screw terminals — OK

12.4 V

### Relay module (5V)

GPIO2 · COM→NO · 12V LED load circuit — OK

CH1 ON · active

### LM2596 Buck converter

12V → 5V · powers ESP32 VIN — OK

5.00V output

## Data pipeline

ESP32 → HTTPS/JSON → Firebase RTDB → App

Dashboard | Forecast | Load | Battery | Hardware

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# Battery Health

12V system · INA219 voltage tracking

60%
SoC

| | |
|---|---|
| Voltage | 12.95 V |
| Current in | +2.00 A |
| Power | 27.27 W |
| Grid voltage | 11.8 V |
| Health | Moderate |

| 12.95V | 34°C | 142 |
|---|---|---|
| Voltage | Temperature | Cycles |
| Normal | Normal | Estimated |

Active alerts · from sensor readings

**High discharge**

INA219: 2000mA – check load

14 Apr 21:28

**Over-discharge protection**

Relay auto-shed activates below 20% via ESP32 logic

**Maintenance reminder**

Check LM2596 output and ESP32 connections monthly

**INA219 calibration**

Dashboard Forecast Load Battery Hardware

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# Load Control

● Relay module · GPIO2 · Smart switching

**Charging hub (CH3)** Medium

Auto ON when battery > 60%

**Fan (CH4)** Optional

Optional · disabled below 50%

Manual override

Force ON | Force OFF

System normal

Battery 60% — all loads available. CH1 relay is ON.

Live power readings

27.7W 2000mA 1952mA 60%

Power INA219 ACS712 Battery

How relay command works

1. Toggle switch → app writes to Firebase relay_command/state
2. ESP32 polls Firebase every 10s → reads new state
3. ESP32 sets GPIO2 HIGH or LOW
4. Relay coil → COM connects to NO terminal
5. 12V flows: ACS712 IP+ → IP- → relay → LED load

Dashboard | Forecast | Load | Battery | Hardware

9:27 79%

# Dashboard

Demo · Updates every 3s

Logout

78%

Moderate

13.20V · Grid 12.6V

1.98A in | -62dBm | Up 47s

0% Estimated from 13.20V INA219 reading 100%

Relay control · GPIO2

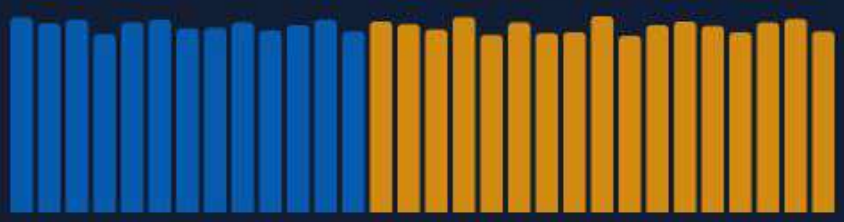
## LED load switch

ON — load is consuming power

Writes to Firebase relay_command/state

Power history · 55 readings

Power (W) Peak 28.3W · Avg 27.0W



Older — Relay ON — Relay OFF Latest

System alerts

High current 1976mA

Last updated: 21:27:14

Dashboard | Forecast | Load | Battery | Hardware

9:27 79%

# Dashboard

Demo · Updates every 3s

Logout

Demo mode · Simulated solar data · Connect ESP32 for live readings

| 3.9 Wh | 28.3 W | 78% |
|---|---|---|
| Today | Peak | Battery |

Live readings · INA219 sensor

Solar output
26.0 W
↑ 51% of max

Current
1959 mA
1.959 A

Voltage
13.35 V
Charging

ACS712
1907 mA
GPIO35 · load current

Battery status

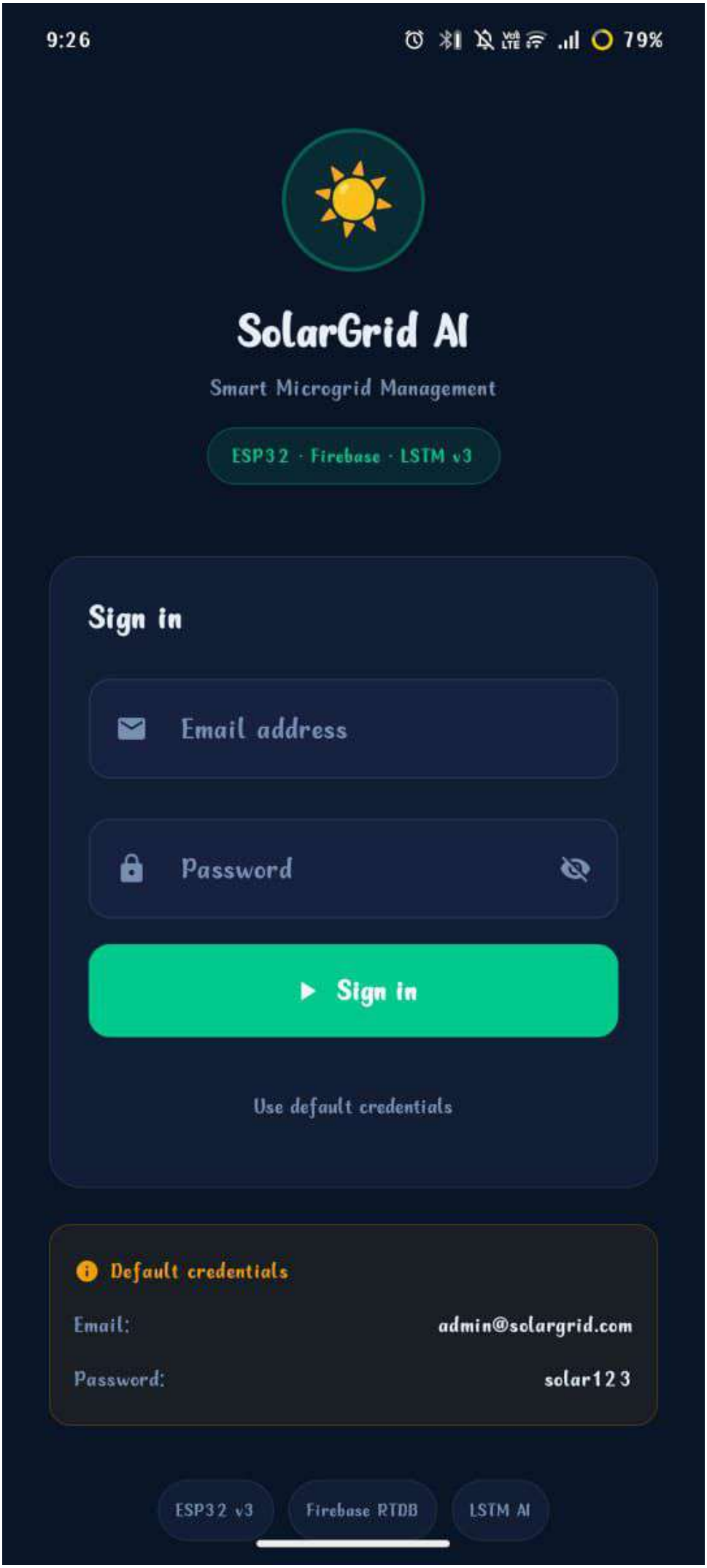
78%

Moderate
13.35V · Grid 12.2V

1.96A in | -64dBm | Up 37s

0% Estimated from 13.35V INA219 reading 100%

Dashboard | Forecast | Load | Battery | Hardware


9:26
79%
SolarGrid AI
Smart Microgrid Management
ESP32 · Firebase · LSTM v3
Sign in
Email address
Password
Sign in
Use default credentials
Default credentials
Email:
admin@solargrid.com
Password:
solar123
ESP32 v3
Firebase RTDB
LSTM AI